

# Cosets and Lagrange's Theorem.

Defn: Let  $G$  be a group,  $H$  a subgroup of  $G$ , for  $a, b \in G$ , we say  $a$  is congruent to  $b \pmod H$ , written as  $a \equiv b \pmod H$  if  $ab^{-1} \in H$ .

Lemma:

Defn: For all  $a \in G$ ,

$$Ha = \{x \in G \mid a \equiv x \pmod H\}$$

where  $Ha = \{ha \mid h \in H\}$ . ( $Ha$  is called the right coset of  $H$  in  $G$ ).

Pf: For shorthand notation, let  $[a] = \{x \in G \mid a \equiv x \pmod H\}$ . We want to show that  $Ha = [a]$ . Towards that, it is sufficient to prove both 1.  $Ha \subseteq [a]$  and 2.  $[a] \subseteq Ha$ .

First part:  $Ha \subseteq [a]$ .

If  $h \in H$ , then  $a(\underbrace{ha})^{-1} = a(\bar{a}^{-1}h^{-1}) = h^{-1} \in H$ .

Here  $a \equiv (ha) \pmod H$ .

Since  $a \equiv ha \pmod H$ ,  $ha \in [a]$ . This is true for all choices of  $h \in H$ .

Second part:  $[a] \subseteq Ha$ .

$x \in [a]$  if  $a \equiv x \pmod H$ . That is,  $ax^{-1} \in H$ .

Further,  $x \equiv a \pmod H \Rightarrow xa^{-1} \in H$ . That is  $xa^{-1} = h$

for some  $h \in H$ .  $\Rightarrow x = ha$  and thus  $x \in Ha$ .

Lemma: 2. There is a one-to-one correspondence between any two right cosets of  $H$  in  $G$ .

1. Any right cosets of  $H$  in  $G$  either are identical or have no element in common.

Pf: Let  $H_a$  and  $H_b$  be two right cosets of  $H$  in  $G$  through the elements  $a, b \in G$ .

Let us assume that  $H_a \cap H_b$  is non-empty. That is, there is an element in both  $H_a$  and  $H_b$ .

Let  $c$  be such an element with  $c = h_1 a = h_2 b$  for some  $h_1, h_2 \in H$ . Now we will show that  $H_a = H_b$  by first showing  $H_a \subseteq H_b$  and then  $H_b \subseteq H_a$ .

First part:  $H_a \subseteq H_b$ .

Since  $h_1 a = h_2 b$ ,  $a = h_1^{-1} h_2 b$ .

Thus for any  $h \in H$ ,  $ha = h(h_1^{-1} h_2 b)$

$$= (h h_1^{-1} h_2) b.$$

Note that  $h h_1^{-1} h_2 \in H$  and thus  $(h h_1^{-1} h_2) b \in H_b$ .

Second part:  $H_b \subseteq H_a$ .

Symmetric argument as above.

$$b = h_2^{-1} h_1 a, \quad hb = h(h_2^{-1} h_1 a) = (h h_2^{-1} h_1) a \in H_a.$$

Define a bijection  $\phi: H_a \rightarrow H_b$  as follows.

$$\phi(ha) = hb.$$

•  $\phi$  is well-defined.

$$ha = h'a \Rightarrow h = h'$$

Injective: If  $\phi(h_1a) = \phi(h_2a)$  then  $h_1b = h_2b$ .

$$\Rightarrow h_1 = h_2$$

Surjective: For any element  $hb \in H_b$ , then there exists  $ha \in H_a$  s.t.  $\phi(ha) = hb$ .

### Lagrange's

Theorem: If  $G$  is a finite group and  $H$  is a subgroup of  $G$ , then  $|H|$  is a divisor of  $|G|$ .

Defn: If  $H$  is a subgroup of  $G$ , the index of  $H$  in  $G$  is the no. of distinct right cosets of  $H$  in  $G$ .

$$i_G(H) = \frac{|G|}{|H|} = \frac{|G|}{|H|}$$

Remark: Every element in  $G$  belongs to a right coset of  $H$  in  $G$ .  
Trivially  $e \in H$  and  $g \in Hg \forall g \in G$ .

Other way to see this: Let  $H = \{h_1, \dots, h_r\}$ .  
 Suppose that  $H \neq G$ . (If  $H = G$ , then it is a trivial case)

Enumerating the elems of  $H$ , we have

Row 1  $h_1, h_2, \dots, h_{r-1}, h_r$

Row 2  $h_1 a, h_2 a, \dots, h_{r-1} a, h_r a$  ← for  $a \in G \setminus H$ ,

for  $b \in G \setminus (H \cup Ha)$ ,

Row 3  $h_1 b, h_2 b, \dots, h_{r-1} b, h_r b$

Claim: None of the elements repeat across the three rows. (nor in the same row).

$$\begin{aligned} h_i a = h_j a &\Rightarrow h_i = h_j \text{ nor } h_i a = h_j \Rightarrow a = h_i^{-1} h_j \\ &\Rightarrow a \in H \text{ (contradiction)} \end{aligned}$$

if  $h_i b$  was in Row 1 or 2 then

$h_i b = h_j \Rightarrow b = h_i^{-1} h_j$  and this implies  $b \in H$  contradicting the fact that  $b \in G \setminus (H \cup Ha)$ .

$h_i b = h_j a \Rightarrow b = h_i^{-1} h_j a = (h_i^{-1} h_j) a \in Ha$ .  
 Again a contradiction.

Repeating this argument until exhaustion of elements, we may write  $k$  rows. Each row corresponds to a distinct coset.  $k \cdot |H| = |G| \Rightarrow k = \frac{|G|}{|H|}$ .

Remark: If  $m$  divides  $|G|$  then  $G$  need not have a subgroup of order  $m$ .

Group  $S_3 = \{e, (12), (13), (23), (123), (132)\}$

$$H = \{e, (123), (132)\}.$$

$$\sigma \equiv \tau \pmod{H} \text{ if } \sigma\tau^{-1} \in H.$$

Ex:  $(12) \equiv (13) \pmod{H}.$

$$\hookrightarrow (12)(13)^{-1} = (12)(13)$$

$$\begin{array}{ccc} 1 \rightarrow 2 & \left( \begin{array}{c} 1 \rightarrow 3 \\ 3 \rightarrow 1 \\ 2 \rightarrow 2 \end{array} \right) & \rightarrow \begin{array}{c} 1 \rightarrow 3 \\ 2 \rightarrow 1 \\ 3 \rightarrow 2 \end{array} \quad (132). \\ 2 \rightarrow 1 & & \\ 3 \rightarrow 3 & & \end{array}$$

$(23) \equiv (12) \pmod{H}$

$$\hookrightarrow (23)(12)^{-1} \in H?$$

$$\left( \begin{array}{c} 1 \rightarrow 2 \\ 2 \rightarrow 1 \\ 3 \rightarrow 3 \end{array} \right)^{-1} \rightarrow \left( \begin{array}{c} 1 \rightarrow 1 \\ 2 \rightarrow 3 \\ 3 \rightarrow 2 \end{array} \right) \rightarrow \begin{array}{c} 1 \rightarrow 3 \\ 2 \rightarrow 1 \\ 3 \rightarrow 2 \end{array} \quad \begin{array}{c} \checkmark \\ \cancel{(122)} \\ (132) \end{array}$$

$e \equiv (123) \pmod{H}$

$$e \cdot (123)^{-1}$$

$$\left( \begin{array}{c} 1 \rightarrow 2 \\ 2 \rightarrow 3 \\ 3 \rightarrow 1 \end{array} \right)^{-1} \rightarrow e = \begin{array}{c} 1 \rightarrow 3 \\ 3 \rightarrow 2 \\ 2 \rightarrow 1 \end{array} \quad (132).$$

Cosets

$$[e] = H \text{ it self}$$

$$[(12)] = \{(12), (13), (23)\}.$$